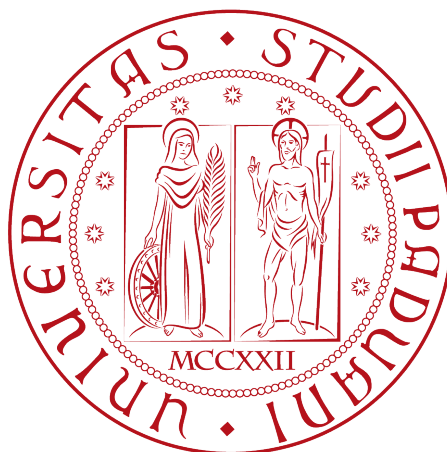


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MASTER DEGREE IN COMPUTER SCIENCE



# GeoMedia: A Geographic Framework for Location-Based Content Sharing

*Master's Thesis*

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# Summary

Over the past few decades, the use of mobile devices has increased drastically, almost replacing desktop software and web browsers. This shift has been driven not only by the widespread adoption of smartphones, but also by the availability of integrated sensors such as GPS, gyroscopes, and Bluetooth, which allows a richer and more context-aware user experiences. By leveraging these technologies, Online Social Media platforms have been able to provide more advanced and sophisticated features; consequently collecting larger amounts of metadata, interaction and daily engagement.

This thesis presents “GeoMedia”, an open-source framework designed for sharing multimedia content — such as text, images, audio, and files — anchored to real-world geographic locations. By leveraging GPS data available on mobile devices, the application provides an interactive map through which users can explore and publish digital content, creating a seamless integration between the digital and physical worlds. Moreover, the framework offers several features and cover various use cases, including restricting content visibility within customizable geographic areas, enabling time-sensitive content, applying user-based restrictions for content creation and visualization, and supporting sequential visibility between published posts. Furthermore, analyzes the motivations behind GeoMedia, its critical aspects and design considerations, describes its system architecture and implementation through a comparison of different technological paradigms. Finally, it explores potential applications and outlines possible future developments of the framework.

An previous version of this framework has been presented at “GoodIT 2025: International Conference on Information Technology for Social Good” (September 3 - 5, 2025 - Antwerp, Belgium) [[paper:goodit](#)]

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# Chapter 1

## Introduction

### 1.1 Overview

Technology has always influenced society, from the first introduction to the Personal Computer to the Internet, changing the world year after year, from personal life to workspaces, people are surrounded by software on their computers and mobile devices. Over the last decades, the investments in the development of technological innovations, both in hardware and software, as increased exponentially, shaping economical and social impact by integrating the latest features into everyday life. Smartphones and tablets have been the core engine that completely changed daily habits and digital usage; their wide adoption has reached almost three quarters of global population [37, 34], making users abandon traditional computers, not just for web browsing but more especially for social media[33] and instant messaging platform[35], that changed the design and development of applications into a ‘mobile first’ logic.

Compared to computers, mobile devices offer distinct functionalities and strengths that have facilitated their widespread adoption. These can be categorized according to the fundamental logic underlying any technology:

1. **Hardware:** the increase in computational performance[23], battery life[6], fast connectivity, and the integration of several built-in sensors[22] such as gyroscopes, GPS, NFC, light sensors, cameras, and vibration motors. These enable applications to interact with different fields such as health[4], education[28], and, through motion-sensors or cameras, an interaction with physical world, creating more immersive user experience.
2. **Software:** the development and presence of mobile applications has played a crucial role. In fact, platforms such as ‘Windows Phone’ were eventually abandoned and discontinued due to the lack of applications [5, 38] compared to operating systems like ‘iOS’[10] and ‘Android’[11].

In this way, smartphones apps have become a key integration with daily life and physical world, for example, when planning a trip or just driving, smartphones, integrated with cars assists users via app like ‘Google Maps’[13] or ‘Waze’[8], providing not just directions but also real-time traffic updates, read events or information about available parking. This makes the surrounding environment enriched with digital details. Another example is ‘Google Lens’[12], which can identify objects from a photograph, or ‘Shazam’[32] that through devices’ microphone can recognize a playing song. Even in workplaces or public areas, technology integrations can be present, from the restaurant that provide a digital menu by a tablet, to customers checking the nutrition macros into a grocery store by scanning the barcode on the pack, to museums by scanning a QR Code enabling audio-guided tours or unlock Augmented Reality functionalities.

## 1.2 Online Social networks

Alongside the widespread adoption of smartphones and mobile devices, online social networks, have emerged as one of the most influential and important sectors of the technology market. Today, social media platforms rank among the firsts most used applications worldwide [29, 9], while the companies that provide them become some of the most bigger actors in global financial markets [30, 19, 16].

Online Social Networks (OSN) are the digitalized form of a Social Network (typically represented as a graph of people where links correspond to relationships) through the Internet[31]. For instance, Facebook, one of the largest platforms, represents a graph of friendships, where nodes (users) are connected via undirected edges, assuming that friendships are mutual. There are also different types of OSN, such as Instagram or Twitter, where the links between users are directed, implementing a “following” model.

These services provide a variety of functionalities, first of all instant messaging capabilities, both one-to-one and one-to-many, and, in addition, the ability to share a wide range of multimedia content, including images, videos, audio recordings, and other types of documents. This facility and advantages compared to other communication forms, like SMS, has led to their integration across diverse social, professional, and public contexts. The particularity of sharing multimedia content, commonly referred to as a ‘post’, has coined the term Social Media, where the published content represent the node of interaction between users through comments or reactions (e.g., ‘like’ or ‘unlike’), thus to replicate human behavior in dialogues and social interactions.

Despite the significant benefits offered by social media services, including enhanced connectivity, community creation, and information sharing, their adoption has also introduced various social and ethical challenges. Concerns regarding user privacy are particularly relevant, as it is often not respected by the service providers themselves, who collect personal information and interests in order to monetize them or enable surveillance practices[20]. Other serious issues include the dissemination of misinformation and fake news, nowadays further amplified by Artificial Intelligence-generated content, commonly known as ‘deepfakes’ [21]. Such content can influence public opinion and social behavior, not always in a positive way [36, 24].

Online Social Networks have also intensified common societal problems through phenomena such as ‘echo chambers’[3], where users are repeatedly exposed to information and content that aligns with their existing beliefs and preferences. This continuous reinforcement can strengthen existing views, reduce exposure to alternative perspectives, and, in some cases, contribute to ideological radicalization. In practice, examples can be observed in so-called ‘incel’ communities [25] and among supporters of extreme political movements.

To address these issues, social media platforms have adopted various moderation and safety mechanisms, including automated content analysis systems, community guidelines and reporting tools. While these measures contribute reducing the spread of harmful content, they are not always fully effective, due to the scale and complexity of emerging innovations, such as ‘deepfake’ [2]. Consequently user reporting and human moderation aside of automated tools, continue to play a crucial role in identifying, evaluating, and removing content that violates policies or legal regulations[7].

## Chapter 2

# Related works

GeoMedia positions itself among existing online social media platforms, offering core functionality for sharing location-based content anchored to geographic coordinates. Content is visible only within a user-defined distance, or it can be used to create various real-world scenarios, such as scavenger hunts or city tour guides. The application enables users to explore the real world through enriched map-based content contributed by other users, who can upload pictures or other multimedia files associated with specific locations.

Several other social media platforms utilize a mobile device's GPS module to enhance user experience or provide additional functionality. However, none fully implement the concept behind GeoMedia or its versatility in creating various application scenarios and explore content uploaded.

### 2.1 Foursquare Swarm

Foursquare Swarm is the spin-off of a formerly app 'Foursquare City Guide', which once provided personalized recommendations of places near a user's current location based on its interests or previous browsing history and check-in (place where the user has been). Foursquare has been discontinued in 2024 and its functionalities have been integrated into Swarm, another app that also allows users to share their position with relatives. The apps also create a personal lifelog of places visited thus to share it with others.

Any user for each place visited, can leave a review or uploads media, but this is associated to a physical place like a store, coffee, restaurant or something, cannot be done in desired coordination and do not provide any kind of post confining, every user can access to it. The application also incentivizes the usage by making the user earning badges by visiting places and uploading content to it. These functionalities and mechanisms are quite the same as also other solutions like Google Maps (see Section 2.4).

#### 2.1.1 Dodgeball and Google Latitude

Dodgeball later acquired by Google and rebranded as 'Google Latitude' [15] was quite the precursor of Foursquare, simply allowing to share current locations with friends through a map. Google Latitude was discontinued in 2013, integrated into another discontinued social media by Google (Google+ [14]) and finally migrated into Google Maps in 2018.

## 2.2 Snapchat

Snapchat[18] app has been one of the most used social media world wide, mainly thanks to its instant-message service and multimedia functionality like time-based content, that make the uploaded content visible only for 24h or visible only once by a user. This brings user to check it daily in order to not missing out anything and to keep up with realtives life. The app also consent to see firends current location on the app, and geographic group closer content into a single flow of information. This service has been a main application of ‘mobile first’ concept for social media, not only leveraging GPS module, but also camera thus to implement augmented reality objects.

Copompared to GeoMedia, the users can explore the information uploaded around them but not categorized on specific interests, more like a bunch of media together.

Despite the innovative functionality this create a negative social impact on daily life, called FoMo (fear of Missing Out). Another problem concerns user privacy, expecially

## 2.3 Instagram

## 2.4 Google Maps

## 2.5 Jagat

## 2.6 Others

- auglinn - wallame?? - strava

## Chapter 3

# GeoMedia

### 3.1 Idea

### 3.2 Functionalities

## Chapter 4

# Architecture and Software design

## Chapter 5

# Mobile app

## Chapter 6

# Geomedia Backend

## Chapter 7

# Conclusions

7.1 Consuntivo finale

7.2 Raggiungimento degli obiettivi

7.3 Conoscenze acquisite

7.4 Valutazione personale

*“Life is really simple, but we insist on making it complicated”*

— Confucius

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*Innanzitutto, vorrei esprimere la mia gratitudine al Prof. Claudio Enrico Palazzi, relatore della mia tesi, per l'aiuto e il sostegno fornitomi durante la stesura del lavoro.*

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*Padua, May 2026*

Alberto Morini

Appendix A

Appendice A

Citazione

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Autore della citazione

# Acronyms and abbreviations

**API** [Application Program Interface](#). 13

**UML** [Unified Modeling Language](#). 13

# Glossary

**API** in informatica con il termine *Application Programming Interface API* (ing. interfaccia di programmazione di un'applicazione) si indica ogni insieme di procedure disponibili al programmatore, di solito raggruppate a formare un set di strumenti specifici per l'espletamento di un determinato compito all'interno di un certo programma. La finalità è ottenere un'astrazione, di solito tra l'hardware e il programmatore o tra software a basso e quello ad alto livello semplificando così il lavoro di programmazione. [12](#)

**UML** in ingegneria del software *UML, Unified Modeling Language* (ing. linguaggio di modellazione unificato) è un linguaggio di modellazione e specifica basato sul paradigma object-oriented. L'*UML* svolge un'importantissima funzione di “lingua franca” nella comunità della progettazione e programmazione a oggetti. Gran parte della letteratura di settore usa tale linguaggio per descrivere soluzioni analitiche e progettuali in modo sintetico e comprensibile a un vasto pubblico. [12](#)

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